

SMART CLASSROOM DOOR STATUS AND TIMER SYSTEM REPORT

1. Project Description, Purpose, and Materials Used

Description and Purpose: The main purpose of this project is to develop a hardware-based up-counting timer system that visually indicates whether a classroom is occupied or empty in university corridors, operating automatically when a class begins. The system aims to eliminate the need for incoming students or instructors to manually check the door. The project is designed to cover a full class duration of 60 minutes. It has been physically implemented entirely using 74xx series digital logic gates, without the use of any microcontrollers or software.

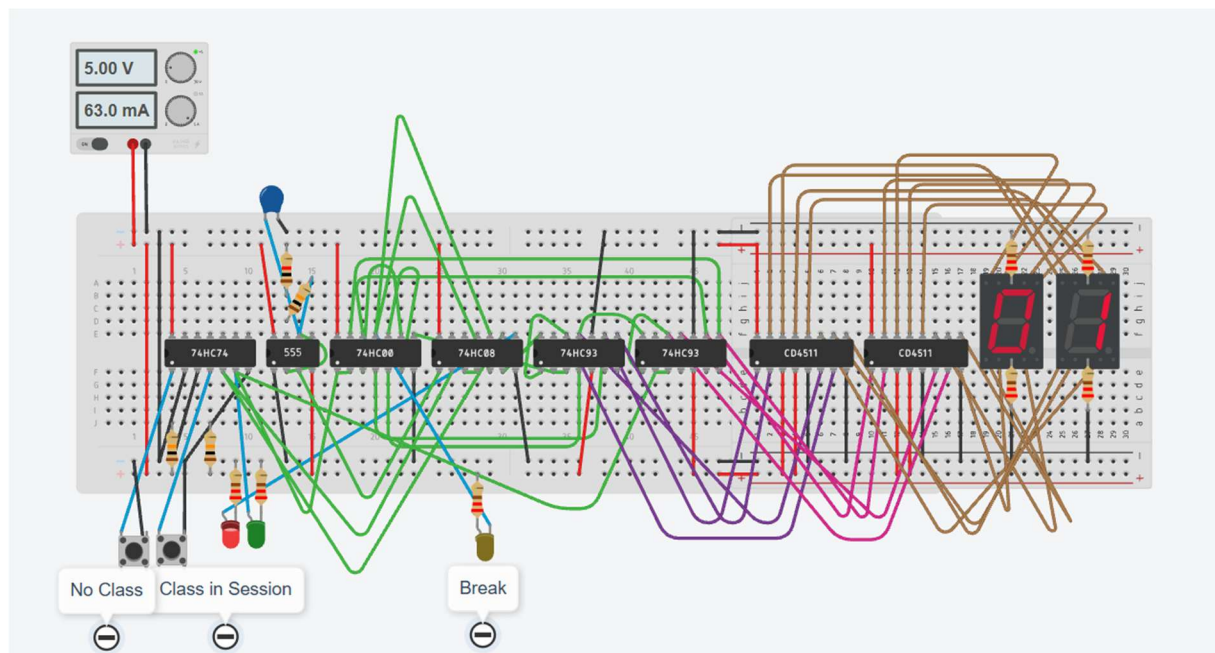
- When the "Class in Session" (Ders Var) button is pressed, the Red LED turns on and the 60-minute countdown begins.
- When the time reaches 60, the system automatically stops counting, the Red LED turns off, and the Yellow LED (Break) turns on.
- When the "No Class" (Ders Yok) button is pressed, the system resets via hardware, and the Green LED turns on, indicating that the classroom is empty.

Materials Used (Physical Prototype):

- **74HC74:** D Flip-Flop (System Memory)
- **NE555 Timer:** Clock signal generator
- **74LS90 (2 Pieces):** BCD Decade Counter (For ones and tens digits)
- **74LS47 (2 Pieces):** BCD to 7-Segment Decoder/Driver (For Common Anode)
- **74HC08:** AND Gate
- **74HC00:** NAND Gate
- **7-Segment Display (2 Pieces):** Common Anode display
- **LEDs:** 1 Red, 1 Green, 1 Yellow
- **Passive Components:** 4700 uF Polarized Capacitor, 100k Multiturn Trimpot, 220 Ohm and 10 kOhm resistors.
- **Infrastructure:** 3x AA Battery Holder (4.5V DC), 2 Breadboards, and Jumper wires.

2. Final State of the Project and Step-by-Step Implementation

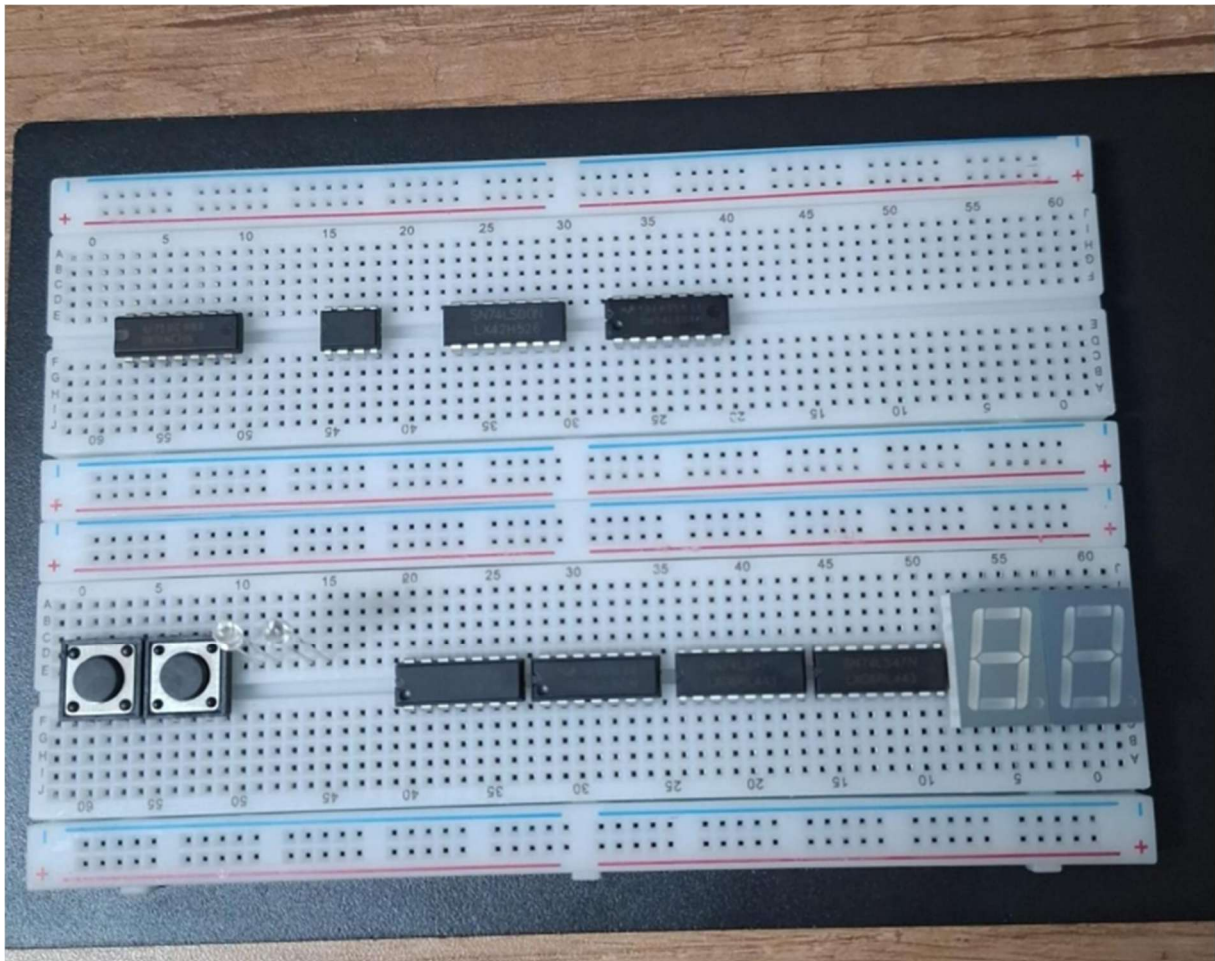
TINKERCAD SIMULATION IMAGE:



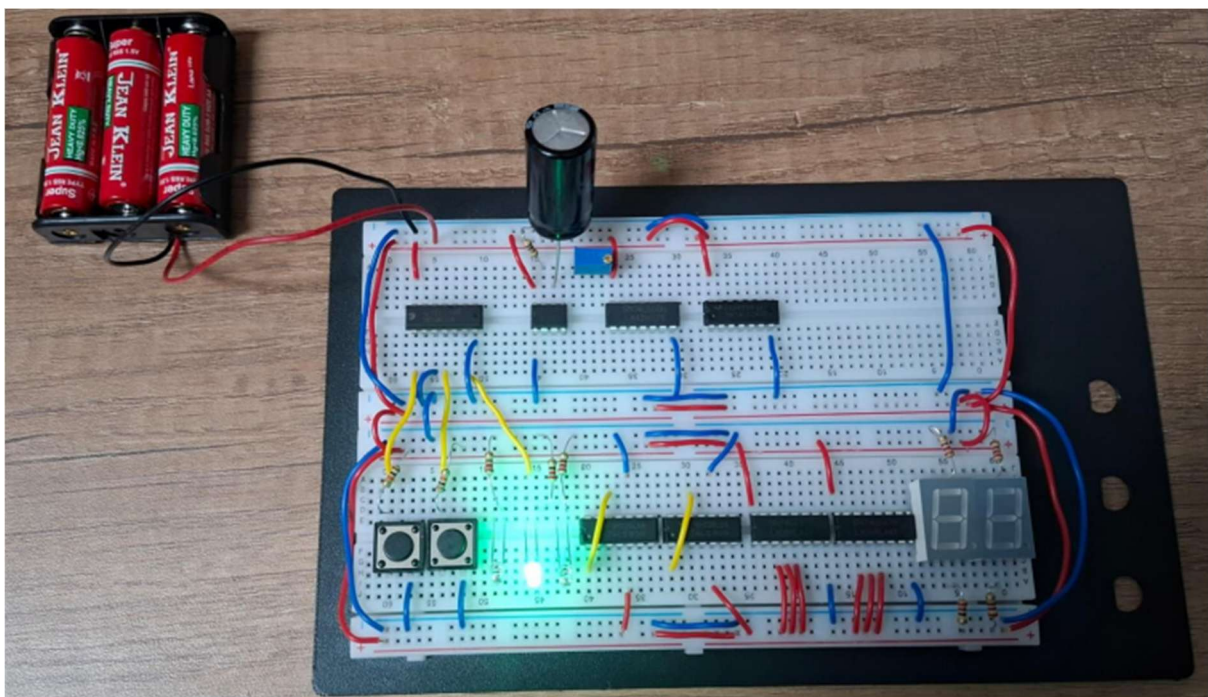
TINKERCAD SIMULATION LINK:

<https://www.tinkercad.com/things/0kYuEpq9gAb-door-availability-system/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard&sha%3Arecode=taiHoGL5doEFiINSHr4JyN1IFtPhtDhkkwLcGUfPFOM>

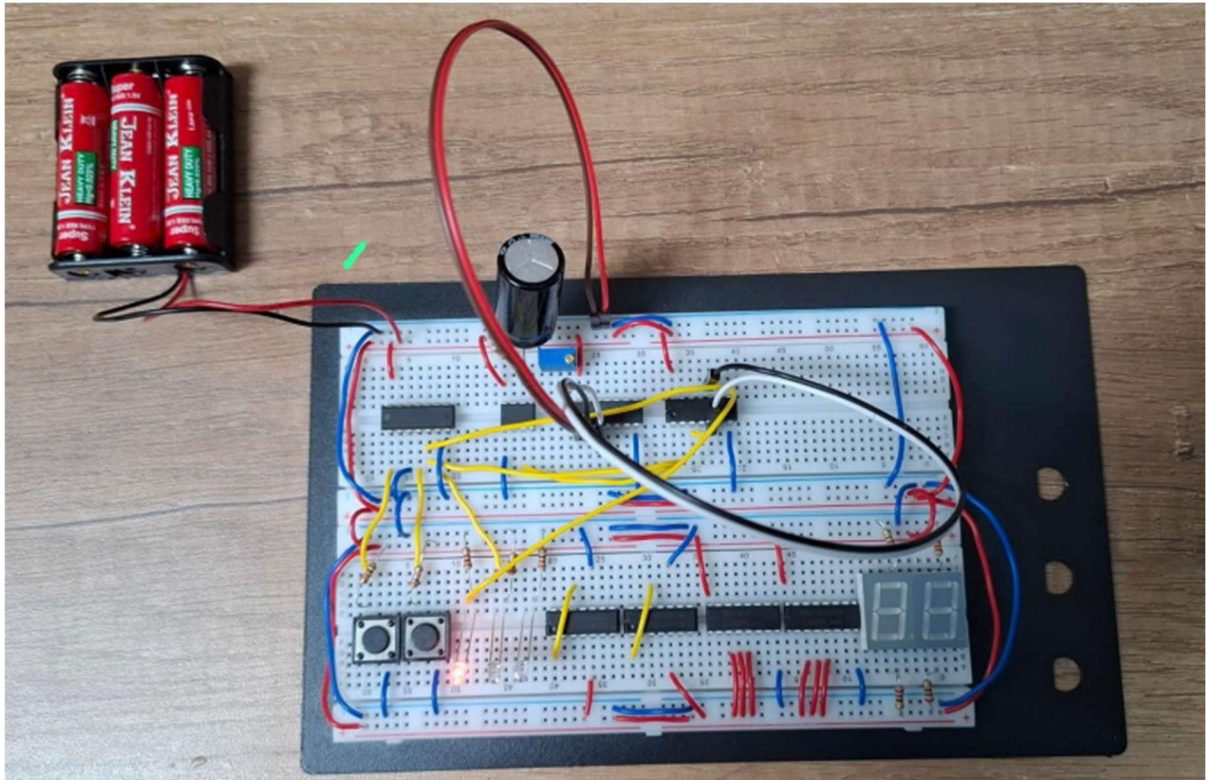
1) Integrated installations:



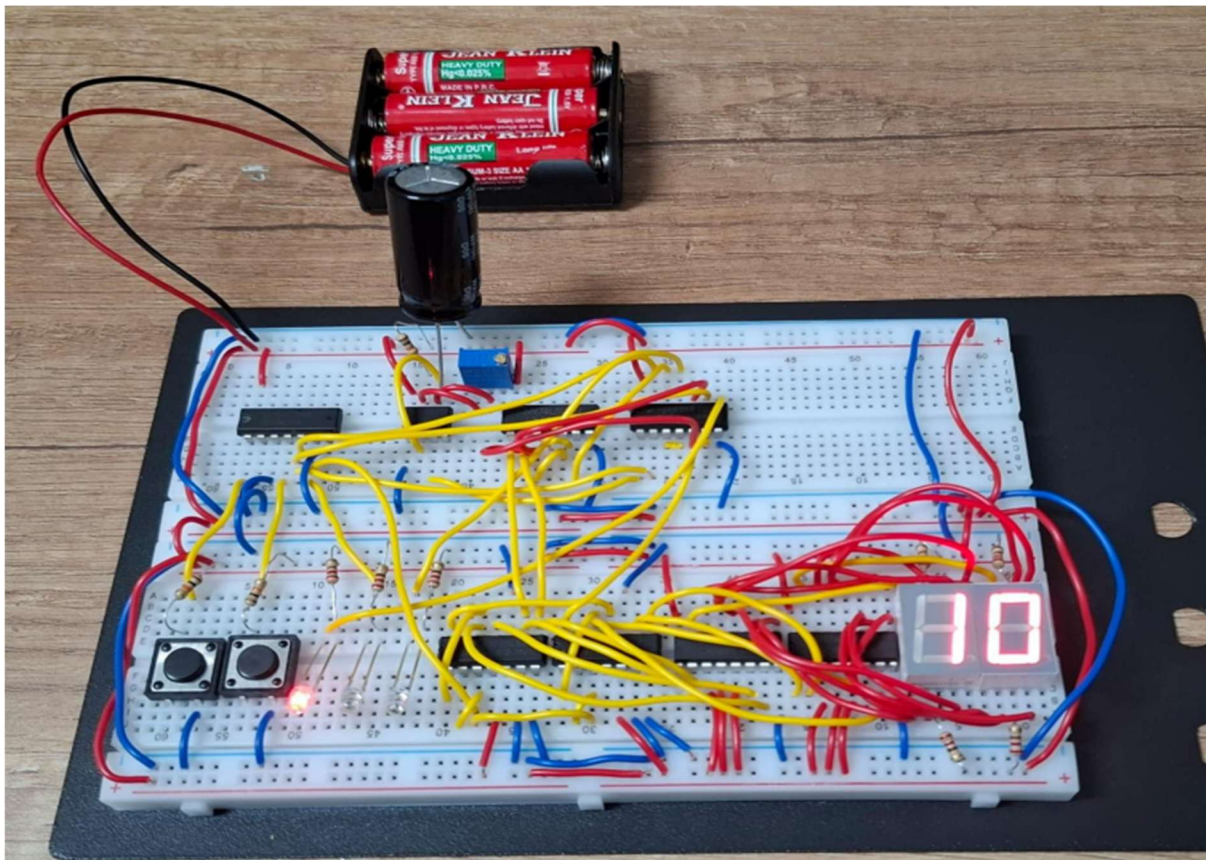
2) Green LED connection:



3) Red LED connection:



4) Display connection:



The project was built in four main steps using a modular breadboard architecture:

- **Step 1: Control Unit and Memory (74HC74):** A 74HC74 Flip-Flop was used for the system to memorize the "Class in Session" or "No Class" states. 10 kOhm Pull-Up resistor networks were established to ensure stability when the buttons are not pressed.
- **Step 2: Clock Signal Generation and Calibration (NE555):** The NE555 integrated circuit was used as the timing unit. To ensure the time increments exactly every 60 seconds, analog calibration was performed using a 4700 uF electrolytic capacitor and a 100k multiturn trimpot, with fine-tuning down to sub-second error margins using a stopwatch.
- **Step 3: Counters and Stop-at-60 Logic (74LS90, 74HC08, 74HC00):** Two 74LS90 ICs, which count directly in BCD, were used as counters. When the count reaches 60 (0110 in binary), the 74HC00 NAND and 74HC08 AND gates detect this state and physically cut off the clock signal, turning off the Red LED and activating the Yellow (Break) LED.
- **Step 4: Display Drivers (74LS47):** The 4-bit binary data from the 74LS90s was transmitted to the 74LS47 driver ICs and visualized on the Common Anode 7-Segment displays.

3. Encountered Problems and Solution Approaches

The hardware engineering problems encountered during the physical setup and testing phases, and the solutions developed for them, are listed below:

- **Problem 1: Driver and Display Incompatibility (Short Circuit Risk)** The unavailability of the originally planned Common Cathode displays and compatible drivers in the market led to a reverse current problem upon direct connection.
- **Solution:** The circuit diagram was updated, Common Anode 7-Segment displays were used, and 74LS47 ICs were integrated into the system to drive these displays, completely redesigning the data path unlike the Tinkercad simulation.
- **Problem 2: Real-Time Duration Calibration** During the conversion of the 1 Hz signal from theoretical simulations into an exact 60.00-second pulse in the physical world, time drifts (speeding up/slowing down) were experienced due to capacitor tolerances.
- **Solution:** Instead of fixed-value resistors, a 100k multiturn trimpot was integrated into the system. The target duration was achieved with 1-second

precision by performing mechanical fine-tuning (calibration) via a screwdriver using real-time stopwatch measurements.